

Is ChatGPT a Boon or a Bane for Learning? Experimental Evidence Across Task Formats and Chatbot Designs

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As concerns mount over the widespread use of large language model-based chatbots for schoolwork, our understanding remains limited regarding their effects on learning and how to customize LLM chatbots and learning tasks to promote effective learning in the era of generative AI. To address these issues, we conducted randomized field experiments among college students, randomly assigning each participant to one of three tools during a practice session: a Bing-based search engine, a standard ChatGPT chatbot, or a custom ChatGPT chatbot designed using the guided discovery principle through zero-shot prompting. Using subsequent paper-and-pencil exams to measure learning outcomes, we found that students with access to standard ChatGPT scored 4.18% lower than those with access to the search engine, despite ChatGPT being 3.8 times more likely to be used. Interestingly, the standard ChatGPT demonstrated negative effects primarily for selected-response problems, whereas positive effects were observed for constructed-response problems. In contrast, students provided with the guided-discovery ChatGPT scored 9.23% higher on exams and outperformed the standard ChatGPT on both selected-response and constructed-response problems, an outcome attributable to extended practice time, more conversation rounds, and greater critical engagement. Finally, post-survey responses revealed that students overestimated the learning benefits of the standard ChatGPT and relied on it more heavily, despite recognizing the advantages of guided-discovery ChatGPT, highlighting the risk of overreliance on standard LLM chatbots.

Key words: large language model, search engine, guided discovery, task format, field experiment

1. Introduction

Since its public launch, ChatGPT and other large language model chatbots (“LLMs” for short) have rapidly permeated educational settings from K-12 classrooms to university lecture halls, offering students on-demand homework support, writing support, and study aids (Borges et al. 2024). Recent surveys report that about 88% of K-12 students, undergraduates, and graduates admit to

using LLMs for schoolwork, with about half using them weekly (Impact Research 2024). Unlike earlier learning technologies studied in the technology-mediated learning literature, which shape learning environments primarily by structuring content, pacing, or motivation (Alavi and Leidner 2001, Bauman and Tuzhilin 2018, Gupta and Bostrom 2013, Ko et al. 2023), LLMs can generate fluent step-by-step solutions, natural-language explanations, and interactive problem-solving guidance. These capabilities enable them to intervene directly in learners’ cognitive processes, effectively acting as on-demand solution generators, reasoning partners, or virtual tutors. This same capability, however, raises important concerns: by performing substantial portions of the cognitive work, LLMs may encourage “cognitive offloading”—the delegation of effortful reasoning to the AI system. Indeed, this fundamental tension between cognitive augmentation and cognitive displacement may explain the divergent views on the roles of LLMs in learning: while many educators, students, and parents view LLMs as valuable learning tools, others worry that reliance on LLMs may encourage shortcut-taking and undermine the development of critical thinking and problem-solving abilities (Bastani et al. 2025, Chien et al. 2025, Lehmann et al. 2024, Lo et al. 2024, Xu et al. 2024).

While public discourse alternates between viewing LLM chatbots as transformative learning aids and as threats to genuine learning and critical thinking, empirical evidence from real-world educational settings remains limited and inconclusive. Reflecting this uncertainty, educational institutions often defer decisions to individual instructors, resulting in wide disparities in restrictions and guidance, ranging from outright bans to full integration of LLMs into assignments and assessments (Freeman 2025). This fragmented response underscores a deeper lack of field-based evidence clarifying whether, when, how, and for whom LLMs enhance learning versus merely encourage cognitive offloading and superficial engagement (An et al. 2025). This evidence gap complicates the development of coherent guidance for LLM use in educational settings and underscores the need for systematic investigation of LLMs in authentic learning contexts.

To best inform educational practice, it is essential to study the effects of LLMs in naturalistic learning settings, where learners retain discretion over whether and how to engage with these tools for real coursework. Equally important is the choice of benchmarks against which LLM effects are evaluated. Prior to the advent of LLMs, search engines had long served as a primary study aid, enabling students to look up information, explore solution approaches, and locate worked examples. These activities overlap with several common ways students use LLMs for learning, making search engines a natural benchmark for LLMs. The evaluation of learning effects of LLMs also depends critically on how these systems are designed and deployed in learning contexts. Learning is not simply a matter of providing answers; in many cases, effective learning requires withholding information or structuring assistance in ways that guide learners to derive solutions on their own.

In their default form, commercial LLMs such as ChatGPT are optimized to produce fluent and helpful responses rather than to promote effective learning. However, LLM-based tools are inherently flexible and can be adapted to educational contexts by incorporating pedagogical principles from learning science that regulate how and when assistance is provided. Accordingly, it is also important to investigate whether alternative LLM designs that integrate pedagogical principles can improve learning outcomes relative to standard, unmodified LLMs.

In addition to system design, a further nuance in understanding the learning effects of LLMs is that they are likely to vary across learning tasks. Learning tasks differ systematically in the cognitive processes they require. For example, selected-response tasks (e.g., multiple-choice questions) emphasize knowledge recognition and retrieval, whereas constructed-response tasks (e.g., problem-solving questions) require solution generation, knowledge integration, and explanation. Because task formats are one of the primary levers instructors can use to influence students' engagement with LLMs, it is important to examine LLM use across different task formats to understand whether and how their effects on learning differ.

Moreover, the effects of LLMs can vary systematically across learner characteristics. For example, differences in prior knowledge or familiarity with LLMs may shape learners' ability to evaluate and build upon LLM-generated responses, thus affecting how these tools influence learning outcomes (Chen and Chan 2024). In addition, prior research on human-AI interaction suggests that demographic differences, including gender, may be associated with variation in the adoption and use of AI systems (Aldasoro et al. 2024), which could in turn lead to differential effects of LLMs on learning outcomes. Accounting for such heterogeneity is essential for developing a nuanced understanding of when and for whom LLMs enhance learning, and for informing instructional decisions about their effective use.

Motivated by these needs, we experimentally investigate three interrelated questions. First, we ask whether students' access to standard LLMs ultimately improves or undermines their learning, relative to search engines. Second, we ask whether incorporating pedagogical principles, specifically guided discovery principles, into standard LLMs enhances their effectiveness as learning tools and, if so, the mechanisms underlying these enhancements. Finally, we assess whether the effects of LLMs vary across task formats, specifically, selected-response versus constructed-response problems, and across learner characteristics, including prior knowledge, experiences with LLMs, and gender.

We thus conducted pre-registered, between-subjects field experiments during 2024-2025 at two universities across 12 courses. Students were randomly assigned to one of three learning tools: a search engine (BING), a standard LLM (BASE-GPT), or a guided-discovery LLM (GD-GPT). The standard LLM replicates the default behavior of OpenAI GPT-4, whereas the guided-discovery LLM operationalizes guided discovery principles (Bruner 1961) by providing hints and conceptual

guidance without presenting direct answers, thereby encouraging students to actively explore and construct their own solutions, rather than relying on ready-made answers (Alfieri et al. 2011, Ozdem-Yilmaz and Bilican 2025). The experiments were embedded in regular class sessions and consisted of five phases: (1) a pre-survey collecting demographic information and baseline measures, (2) an instructor-led lecture, (3) an assisted practice phase where students worked through a series of selected-response and constructed-response problems, with the option of using their assigned tool (BING, BASE-GPT, or GD-GPT), (4) a post-survey gathering evaluations of the learning tool and the study experiences, and (5) learning outcome assessment using a closed-book exam consisting of problems matched to the practice tasks. Overall, 583 students from two experiments completed a total of 7,695 practice problems and 7,831 exam problems.

Results show that students who were assigned to BASE-GPT underperformed those assigned to BING on the exam, suggesting that access to standard LLMs can impair learning. In contrast, students assigned to GD-GPT outperformed those assigned to BING on the exam, underscoring the effectiveness of the guided-discovery design. We further find that the impact of BASE-GPT on learning varies across task formats. Relative to BING, access to BASE-GPT reduced exam performance on selected-response problems but increased exam performance on constructive-response problems. GD-GPT significantly outperformed both BASE-GPT and BING on both selected-response and constructed-response problems.

Granular analyses of tool usage uncover the mechanisms through which GD-GPT enhanced learning. First, GD-GPT increased both learning time and efficiency. Students using GD-GPT dedicated more time to engage with practice materials and demonstrated a stronger marginal return of practice time on learning performance. Second, GD-GPT prompted more follow-up questions and a greater number of conversation turns, indicating deeper engagement with LLMs. Third, students using GD-GPT more critically engaged in learning and submitted answers that diverged more from the LLM’s outputs, as they selectively rejected more LLM suggestions and incorporated more original ideas not present in the LLM’s responses. These epistemically independent behaviors indicated greater critical engagement when using GD-GPT. Mediation analyses show that learning time, depth of engagement, and critical engagement, each contribute to explaining the observed improvements in exam performance.

Heterogeneity analyses reveal that the guided-discovery design reversed the gender gap in LLM usage. While female students were 10.6% less likely to use BASE-GPT than males, they were 8.7% more likely to use GD-GPT. Female students also benefited more from the guided-discovery design, gaining 7.38 additional percentage points in exam performance compared to male students using the guided-discovery LLM. Together, these differences suggest that female students were more receptive to the guided-discovery design and derived greater learning benefits from it. Furthermore,

students with more LLM experience and stronger prior knowledge derive greater benefits from GD-GPT, suggesting that the guided-discovery design becomes increasingly effective as students’ familiarity with LLMs and the subject domain grows.

Post-survey highlights some discrepancies between students’ perceptions and their actual behaviors and learning outcomes. Despite BASE-GPT’s negative impact on learning outcomes, students rated it highly on germane cognitive load, which is typically associated with deeper learning. This discrepancy suggests that students overestimated the standard LLM’s effectiveness as a learning tool, consistent with the “illusion of learning.” Similarly, although students were 12.9% less likely to use GD-GPT than BASE-GPT in the field, they perceived GD-GPT as the more helpful tool and reported higher intentions to use it. This paradox suggests difficulties in self-regulating LLM use for learning, leading students to over-reliance on standard LLMs that are more convenient but less effective at promoting genuine learning.

This study advances the emerging literature on LLM-supported learning by showing that the effects of LLMs for learning are contingent rather than inherent, and by identifying two critical determinants of their impact: learning task format and human-LLM interaction design. Second, it demonstrates that core pedagogical principles, such as guided discovery, can be encoded directly into LLMs through zero-shot prompting, offering a successful, scalable, and theoretically grounded approach for operationalizing educational theory in LLM-supported learning environments. Third, this study advances guided discovery theory by unpacking its underlying mechanisms using fine-grained behavioral traces that are rarely accessible in conventional learning environments. We show that learning time, depth of engagement, and critical engagement each contribute to improved learning outcomes under the guided-discovery design, suggesting concrete pathways that future LLM-based learning systems can leverage to enhance learning. Finally, these insights offer practical guidance for educators on designing learning task formats that effectively align with LLMs, and for policymakers in developing guidelines for responsible LLM use, thereby helping ensure that LLMs support, rather than substitute for, genuine learning.

2. Related Literature

2.1. Technology-mediated learning

The information systems field has a long history of studying technology-mediated learning, examining how digital technologies shape learning processes and outcomes (Leidner and Jarvenpaa 1993, Alavi 1994, Alavi and Leidner 2001, Gupta and Bostrom 2013, 2009, Piccoli et al. 2001). A central stream of this literature investigates how technologies can engage learners, promote knowledge retention, and improve learning outcomes (Liu et al. 2017, Hu and Hui 2012, Leidner and Jarvenpaa 1995, Franceschi et al. 2009, Gao et al. 2021). Building on this insight, prior work has

investigated how technology-enabled interventions, such as call-to-action prompts (Huang et al. 2021), on-the-hour cues (Huang et al. 2024), and gamification features (Park et al. 2019, Leung et al. 2023, Santhanam et al. 2016), can be used to shape their learning processes by reducing procrastination (Huang et al. 2021), cultivating divergent thinking (Chen et al. 2025), and supporting self-regulated learning (Santhanam et al. 2008). Related studies further demonstrate how personalization mechanisms can address learners’ knowledge gaps (Bauman and Tuzhilin 2018) and optimize learning content sequencing (Li et al. 2024). Extending this stream of research, our study examines LLM-based chatbots as a new form of technology-mediated learning support, investigating both their effects on learning outcomes and how their design can be structured to better support effective learning processes.

In the broader literature, researchers have studied early use cases for rule- or intent-based chatbots with predefined dialog flows (Ko et al. 2023). Early studies of such chatbots have shown that they can improve learning when providing goal-directed feedback (McKendree 1990), emphasizing practical examples (Ahrens and Sankar 1993), generating explanations (Gregor and Benbasat 1999), and supporting error recovery (Sein and Santhanam 1999). However, their reliance on predefined intents and scripted dialog flows constrained the scope and adaptability of interactions, limiting their ability to support deeper learning processes such as sensemaking, interactive problem-solving, and exploratory learning (Han et al. 2025). In contrast, LLM chatbots enable more flexible, context-sensitive interactions by dynamically generating explanations and responses that are unnecessarily pre-specified, thereby expanding the range of learning processes that can be supported. In the next two subsections, we review the nascent literature on the effects of LLM chatbots on learning and other cognitive tasks, as well as emerging studies of customizing LLM chatbots for effective learning.

2.2. Impact of LLMs on Learning and Other Cognitive Tasks

Since the advent of ChatGPT, LLMs have rapidly diffused into students’ learning practices, reshaping how they access information, seek assistance, and engage with learning content (Ross 2025). These models are most commonly deployed as chatbots that provide on-demand learning support for question answering, information retrieval, and learning assessment (Schuetzler et al. 2024). Reflecting this rapid adoption, a growing body of research has begun to examine their impact on learning. For example, recent studies have shown that LLM chatbots can achieve learning outcomes comparable to those of human teachers in English learning tasks (Cheon et al. 2024), and students are more willing to seek help from LLM agents than from human instructors (Gaskin et al. 2024).

Although early evidence points to potential learning benefits of LLMs, their largely unregulated use in educational settings has generated significant concerns. Recent qualitative studies and meta-reviews raise concerns that learners may increasingly favor fast and efficient solutions provided by

LLMs over slower, effortful sensemaking, potentially encouraging shortcut-taking and undermining the development of critical thinking and problem-solving skills (Mogavi et al. 2024, Zhai et al. 2024). Relatedly, empirical evidence indicates that access to LLMs can reduce behavioral engagement with learning materials. For example, Kong et al. (2025) documented shorter note-taking, less time spent studying, and lower instructional video completion among students with LLM access.

Although researchers have begun to unpack the effects of LLMs on learning, the extant evidence remains insufficient to draw definitive conclusions, leaving several important questions unanswered. A portion of the literature relies on self-reported surveys (Essel et al. 2024) or Prolific/MTurk-based online experiments (Kumar et al. 2025, Melumad and Yun 2025). For example, Essel et al. (2024) found that students using ChatGPT reported significant improvements in critical, creative, and reflective thinking. Using online experiments conducted on a crowdsourcing platform, Kumar et al. (2025) showed that adult learners who received LLM-generated explanations performed better on test problems than those who only saw correct answers. In contrast, Melumad and Yun (2025)’s online experiments suggested that participants who learned via LLM summaries developed shallower knowledge compared with those who learned through traditional web search.

More recently, researchers have begun to examine the impact of LLMs in field settings, offering a more realistic assessment of real-world impact. Most notably, Bastani et al. (2025) conducted a field experiment in high-school math classes and found that, relative to providing access to books and notes, providing the standard ChatGPT inhibited math learning as indicated by reduced exam performance. Although their study highlighted potential harms associated with ChatGPT use, it remains unclear why this is the case, and whether LLMs perform better or worse than traditional learning support technologies such as Google and Bing. Moreover, while prior research has documented heterogeneous effects of technology-mediated learning across task formats and learner attributes, including prior knowledge and preparation (Li et al. 2024), gender (Bao et al. 2026), and socioeconomic background (Hansen and Reich 2015), little is known about how LLM effects vary across problem types or learner attributes. This study aims to address some of these gaps and offer new field evidence on the effects of LLMs in higher education.

Our study is also part of a broader, emerging literature that examines how LLMs impact human performance, behaviors, and processes across a range of cognitive and social-cognitive settings. For example, studies have shown that LLMs either enhance or hinder individual creativity in ad writing depending on the user’s expertise level and whether the LLM is used as the main text generator or sounding board (Chen and Chan 2024, Lee and Chung 2024). On the knowledge work platforms, the emergence of LLMs has reduced human voluntary contribution to online programming support platforms (Shan and Qiu 2025) and freelancing job demand for writing and coding tasks (Demirci et al. 2025). There is also a literature on the impact of LLMs on individual beliefs and behaviors

under different settings. For example, dialogues with LLMs are shown to help reduce conspiracy beliefs (Costello et al. 2024), improve online political conversations (Argyle et al. 2023), and enhance human productivity (Noy and Zhang 2023). In group settings, LLMs have been found to facilitate group interactions, help humans find common ground, and reduce polarization (Tessler et al. 2024).

2.3. Customizing LLM Chatbots for Learning

Commercially available LLMs, such as ChatGPT, prioritize fluency by default: they often provide complete and “ready” answers that are easy to consume. However, this may lead to “cognitive offloading”, allowing learners to circumvent the cognitive process involved in problem solving, potentially undermining the learning benefits of LLMs. Recognizing this limitation, a few emerging studies have begun to customize LLM chatbots to enhance their learning benefits.

Many of these efforts draw on the scaffolding principles (Wood et al. 1976), which emphasize adaptive support— adjusting the level of support according to the assessment of learners’ current performance level (contingency), progressive removal of support as learners become more proficient (fading), and enabling learners’ gradual assumption of greater responsibility for task performance (transfer of responsibility) (Van de Pol et al. 2010). For example, Sengewald et al. (2024) introduced a “faded” AI tutor for SQL learning that required students to wait 90 seconds before receiving AI assistance. Bastani et al. (2025) developed a custom “GPT tutor” for high-school math that was preloaded with problem solutions and teachers’ notes and offered step-by-step scaffolding support tailored to students’ progress. Neither the “faded” AI tutor nor the “GPT tutor” improved learning, though the “GPT Tutor” performed better than standard ChatGPT. Chen et al. (2025) designed a Socratic-style LLM that did not provide direct answers but posed personalized questions according to students’ prior knowledge and profiles. They showed that their approach can promote divergent thinking, but only for low-difficulty learning tasks. Ma et al. (2024) designed a Socratic-style LLM that improved the learning performance, but also resulted in lower user satisfaction.¹

Collectively, these studies highlight the necessity and the challenges of incorporating learning theories into LLMs. Several existing designs failed to improve learning outcomes. In addition, some custom LLM chatbot designs, while pedagogically sound, may face adoption challenges due to reduced student satisfaction (Ma et al. 2024). Approaches that rely on the availability of detailed problem solutions or instructor-generated notes may pose implementation challenges when supporting open-ended learning environments (Bastani et al. 2025). Consequently, substantial further research is still needed to explore designs that can meaningfully enhance student learning, while retaining LLMs’ appeal to learners and facilitating large-scale deployment.

¹ It is worth noting that, during the development of our manuscript, a few major LLM providers have rolled out their own “study mode” that more or less follows the scaffolding principles, such as Gemini’s guided learning mode (launched in August 2025), ChatGPT’s study mode (launched in July 2025). However, the effects of these features on student learning are not yet well understood.

3. Design of Custom LLM Chatbots for Learning

Our custom LLM chatbot is intended to support formative learning activities, including practice, assignment-related support, and clarification, rather than to function as a tutoring system. The primary design goals of our custom GPT chatbot are threefold: First, it aims to enhance learning performance by incorporating well-established pedagogical principles that support effective cognitive processes. Second, it is designed to be an appealing learning aid that encourages student engagement. Third, it is intended to be easily scalable and implementable in real-world educational settings with minimal human effort.

3.1. Integrating Guided Discovery Theory into LLMs

Although the Socratic method and scaffolding theory represent two of the most well-established and widely-adopted approaches guiding LLM-based chatbot design, each poses distinct challenges in relation to our specific context and design objectives. First, we aim for a design characterized by high student appeal and adoption. Although Socratic method’s iterative questioning approach offers significant pedagogical benefits for deep learning, prior research has observed reduced student satisfaction in practice support contexts (Ma et al. 2024). Second, our design prioritizes ease of implementation and scalability. As a result, it may not rely on the availability of upfront validated solutions, instructor-generated teaching notes, or information about each student’s knowledge boundaries. Given these considerations, we explored theoretical foundations beyond these two established approaches.

We choose *guided discovery* as our main theoretical framework. The guided discovery principle suggests that, compared to explicit instruction and pure independent discovery, learners benefit more from active exploration with some degree of guidance, such as examples and elicited explanations, without directly providing the target information or conceptual understanding (Alfieri et al. 2011, Mayer 2004, Bruner 1961). Like scaffolding and the Socratic method, the guided-discovery principle is also grounded in the constructivist view of learning, which posits that learning is an active process in which learners actively construct knowledge and discover patterns themselves through interaction with materials, exploration of phenomena, and trials in applying principles (Bada and Olusegun 2015, Bok 2008). While guided discovery can take a variety of forms, it commonly emphasizes providing learners with two types of support: (a) helping them structure a problem by breaking down a (complex) problem into smaller, more manageable steps, and (b) providing the relevant background knowledge or potential direction to support learners’ discovery of the solution.

Theory and empirical evidence attribute guided discovery’s effectiveness to its satisfaction of two critical active learning criteria (Shulman and Keislar 1966, Mayer 2004). First, compared to

conventional instruction that provides full guidance and correct answers, guided discovery better activates appropriate prior knowledge for the active construction of new information. Second, in contrast to pure discovery without guidance, guided discovery facilitates the integration of new information by providing a foundation of appropriate knowledge and freeing up working memory for new knowledge construction (Alfieri et al. 2011).

Guided discovery better aligns with our context and design goals for two main reasons. First, guided discovery does not require adaptive guidance tailored to learners’ knowledge boundaries (Van de Pol et al. 2010, Alfieri et al. 2011), which may not be readily available in our context. Second, guided discovery emphasizes providing structure and background knowledge without giving away final answers or relying on persistent questioning or challenges to learners’ assumptions. Therefore, it may be more aligned with the goal of supporting learners during formative learning activities.

3.2. Implementing the Guided-Discovery Design

Despite the promising potential of guided discovery theory, the design and integration of guided discovery principles into LLM-based chatbots remain largely unexplored. As one of the first studies to design and evaluate a guided-discovery LLM chatbot, we prioritize a scalable and low-cost implementation approach. Specifically, we rely primarily on prompt engineering to embed guided discovery principles into the chatbot design. This prompt-engineering-based approach serves as a proof of concept for guided-discovery LLM chatbots. If shown to be effective, future production deployments may further optimize this design using techniques such as model fine-tuning.

Drawing on guided discovery principles (Alfieri et al. 2011, Ozdem-Yilmaz and Bilican 2025), we developed system prompts that instruct the LLM to guide learners through problem decomposition and the provision of relevant background information, while withholding direct answers. In addition, we adopted a zero-shot design that does not rely on worked solutions, examples, or instructor-generated notes. While integrating expert knowledge, such as worked solutions and instructor notes, can improve the quality of chatbot responses and reduce hallucinations, relying on such knowledge entails two important drawbacks. First, it introduces additional implementation challenges, such as increased financial cost, substantial human effort, and reduced applicability to general-purpose, cross-subject, or open-ended learning contexts. Second, incorporating expert knowledge may confound the effects of the guided-discovery design with those of expert knowledge itself, thereby undermining our goal of isolating the impact of the guided-discovery LLM design.

In sum, we implemented the guided-discovery principles through zero-shot prompting. The full system prompt for the guided discovery chatbot is presented in Appendix A.2.

4. Field Experiment

We conducted pre-registered², between-subject randomized field experiments, in which students were randomly assigned to one of three learning tools with equal probability: a standard LLM chatbot (BASE-GPT), a custom LLM chatbot (GD-GPT), or a search engine (BING, our control condition). The experiment was implemented during selected class sessions across 12 undergraduate and graduate analytics courses over four academic semesters (Spring 2024, Fall 2024, Spring 2025, and Fall 2025) at two Chinese universities.

4.1. Field Experiment Procedures

Each class session, lasting about 1.5 hours, was structured into five phases:

Phase 1. Pre-survey: Students completed a 10-minute survey that collected demographic information, such as the frequency of search engine and LLM (e.g., ChatGPT and DeepSeek) use, and their familiarity with the topics to be covered.

Phase 2. Instructor-led Lecture: The instructor delivered an in-person lecture on the topics outlined in the syllabus.

Phase 3. Assisted Practice (Treatment Phase): Students then participated in a practice phase to reinforce their understanding of the lecture topics. They completed the practice on their laptops using a custom-designed web application that delivered a predetermined set of practice problems created by the course instructor. The practice problems included both selected-response (multiple-choice) and constructed-response (problem-solving) tasks. The application sequentially presented the same set of practice problems to all students and provided the corresponding learning tool based on each student’s experimental condition (see the next subsection for details). Students decided for each practice problem whether to use their assigned learning tool. They were instructed to work independently, without communication with peers or seeking help from instructors. To minimize interference from other learning aids, the application was run in full-screen mode throughout the practice, restricting access to other websites or applications. To ensure compliance, the instructor, researchers, and two research assistants actively monitored the classroom during the experiment.

Phase 4. Post-Survey: After the practice, students completed a post-survey to share their experiences and evaluate the learning tool.

Phase 5. Unassisted Examination: Finally, students took a closed-book, paper-and-pencil exam in class, designed to prevent the use of any learning tools, and proctored by the same personnel who supervised the practice. The exam contained the same number of problems as the practice set, and instructors carefully designed each exam problem to be one-to-one matched with

² This experiment was pre-registered on the Open Science Framework: <https://doi.org/10.17605/OSF.IO/VZ9ER>

a corresponding practice problem, ensuring that each exam-practice problem pair covers the same topic and in the same format. This design enables identification of the tool’s effects at the student-problem level and supports a fine-grained analysis of how human-AI interactions on each practice problem shape learning performance on the matched exam problem, as well as how these effects vary across task formats.

All practice and exam problems were carefully graded. For selected-response problems, the web application automatically evaluated the correctness of students’ answers. For constructed-response problems, three research assistants independently performed blinded grading, and their scores were highly consistent, with the Intraclass Correlation Coefficient (ICC) greater than 0.9; the average of the three grades was used for analysis. Instructors assigned 10% of the course final grade to each experimental session, as specified in the syllabus, with 4% for completing the practice phase and 6% for exam performance, ensuring students took the experiment seriously.

4.2. The System for Assisted Practice

After students logged in to the practice web application using credentials provided by the research team, they first watched a demonstration video about how to use their assigned learning tool (BING, BASE-GPT, or GD-GPT) before they started to work on the practice problems. Each practice problem was shown side by side with the assigned learning tool, whose use was voluntary. Students could choose to use the learning tool to help solve the problem. Upon submitting their answer to a problem, students were immediately shown the correct solution. Then, students could click the “Next” button to proceed to the next problem. They were not permitted to return to previous problems or change their answers. Our web application kept logs of students’ clicks and page views, along with detailed records of tool usage, including search queries and web pages accessed in the BING condition, and conversation histories in the BASE-GPT and GD-GPT conditions.

The system interface consisted of two panels: a practice-problem panel on the left, where students could submit answers, and a learning-tool panel on the right, which varied by the participant’s assigned experimental group. For participants in the BING group, the right panel presented a Microsoft Bing-powered search engine at the top with a search result page below (see Figure EC.1 in Appendix A for a screenshot). Participants in the BASE-GPT and GD-GPT groups were provided access to an LLM chatbot. They could click the “Ask the bot” button at the top of the right panel to automatically paste practice problems into the chat window and receive responses per experimental condition. They could also ask follow-up questions in the chat window (see Figure EC.2 for the screenshot).

BASE-GPT and GD-GPT were both implemented using OpenAI’s GPT-4 API and an interface similar to ChatGPT. They differed only in the system prompts (see Appendix A for details).

BASE-GPT used a minimal prompt to mimic the standard ChatGPT as a learning assistant, whereas GD-GPT was additionally prompted with the guided discovery principle and instructions not to disclose direct answers. For the selected-response problem, BASE-GPT directly provided the correct answer. In contrast, GD-GPT did not, but rather explained the involved terms and suggested ways to think about the problem. Similarly, for the constructed-response problem, BASE-GPT also directly provided the correct answer, while GD-GPT encouraged students to explore the answer by themselves. The conversation logs substantiate that our operationalization effectively aligns with the design objectives.

4.3. Experimental Subjects and Demographics

We began with a pilot study in one analytical course during Spring 2024 to assess system stability and inform power analysis, in which participants were given access to BASE-GPT. Using the pilot data, we estimate that approximately 700 problems per experimental group are required to detect a moderate effect size (Cohen’s $d = 0.15$) with 80% power at a 5% significance level ($SD = 48.4$).

Then, we conducted two field experiments: Experiment 1 was implemented across two academic semesters (Fall 2024 and Spring 2025), and Experiment 2 was conducted in Fall 2025. The two experiments involved a total of 583 students across 11 analytics-related courses. Each experiment comprised multiple sessions, with each session completed within a single class session. Two courses contributed multiple sessions, resulting in a total of 16 course-sessions and 727 student-sessions. In each session, students worked on multiple problems during the practice phase and the same number of problems during the exam phase. In total, two experiments assigned 7,871 problems in each phase. Participants completed 7,695 practice problems and 7,831 exam problems. The difference reflects 10 students who did not complete both phases—excluding them yields consistent results.

Experiment 1 served as our primary study, involving 296 students who completed 3,103 practice problems and 3,239 exam problems. Experiment 2, which included 282 students who completed 4,592 practice problems and 4,592 exam problems, served as a robustness check, replicating the core design of Experiment 1 while introducing a targeted modification to address a potential concern related to the LLM interface design. The design and findings of Experiment 2 are presented in Online Appendix D.2.

Table 1 presents summary statistics for Experiment 1. It shows that 35% of students were randomly assigned to the BASE-GPT group, 32% to the GD-GPT group, and the remaining students to the BING group. Among the participants, 42% self-identified as *Female*, the average *Age* was 22.06 years, and 96% were *Domestic* students. Pre-survey measures captured learning background: *Social Status* was rated on a 1-10 scale (1 = lowest, 10 = highest) with a mean of 6.96, and students reported an average target *Learning Goal* of an A grade. For technology

Table 1 Summary Statistics of Experimental Students

Variable	Description	Mean	SD	Min	Max
Panel A: Student level					
BASE-GPT	=1 if the student was randomly assigned to the BASE-GPT group and 0 otherwise	0.35	0.48	0	1
GD-GPT	=1 if the student was randomly assigned to the GD-GPT group and 0 otherwise	0.32	0.47	0	1
Female	=1 if the student was born female and 0 otherwise	0.42	0.49	0	1
Age	The age of the student	22.06	3.51	17	35
Nation	=1 if the student was a domestic student; =0 if the student was an international student.	0.96	0.19	0	1
Social Status	The students perceived social status, measured from 1(lowest) to 10 (highest).	6.96	1.74	1	10
Learning Goal	The students set their learning goals using the following scale: B (1), B+ (2), A (3), and A+ (4)	3.18	0.78	1	4
Search Engine Use	The frequency of search engine use over the past month, with responses ranging from 0 (Never) to 5 (Daily).	4.36	0.96	1	5
LLM Use	The frequency of LLM chatbot use over the past month, with responses ranging from 0 (Never) to 5 (Daily).	4.02	1.02	1	5
IT Self-efficacy	The students perceived self-efficacy regarding information technology.	3.54	0.83	1	5
Course Familiarity	The students familiarity with the course in general, from not familiar at all (1) to completely familiar (5)	2.20	0.99	1	5
Topic Familiarity	The average value of the students familiarity with each topic covered in this class session, from not familiar at all (1) to completely familiar (5)	2.05	0.88	1	5
Panel B: Student-Problem level					
Constructed-response	=1 if the problem is a constructed-response problem; 0 if a selected-response problem.	0.23	0.42	0	1
Tool Use	=1 if the student used the learning tool for this practice problem; 0 if not.	0.51	0.50	0	1
Practice Time	The minutes a student spent on a practice problem, including time spent using the learning tool.	1.16	1.52	0.01	8.03
Practice Score	The score of the practice problem on a 100-point scale	74.26	39.16	0	100
Exam Score	The score of the exam problem on a 100-point scale	58.83	45.88	0	100

use, students reported frequent *Search Engine Use* and *LLM Use*, and average *IT Self-efficacy* (Santhanam et al. 2008) was 3.54 on a 5-point Likert scale.³ Students also reported relatively low *Course Familiarity* (2.20 out of 5) and *Topic Familiarity* (2.05 out of 5), both measured on a 1-5 scale (1 = not familiar at all, 5 = completely familiar). Panel B summarizes students’ practice and exam performances. Among all the assigned problems, 23% were constructed-response problems. For ease of comparison, all problem scores were scaled to a 100-point scale. On practice problems,

³ Adapted from Santhanam et al. (2008), IT Self-efficacy is measured as the average of the following three items: (1). I feel confident using information technology to learn about and apply new concepts. (2). I would be comfortable using information technology to learn about and apply new concepts. (3). I could successfully use information technology to learn about and apply new concepts.

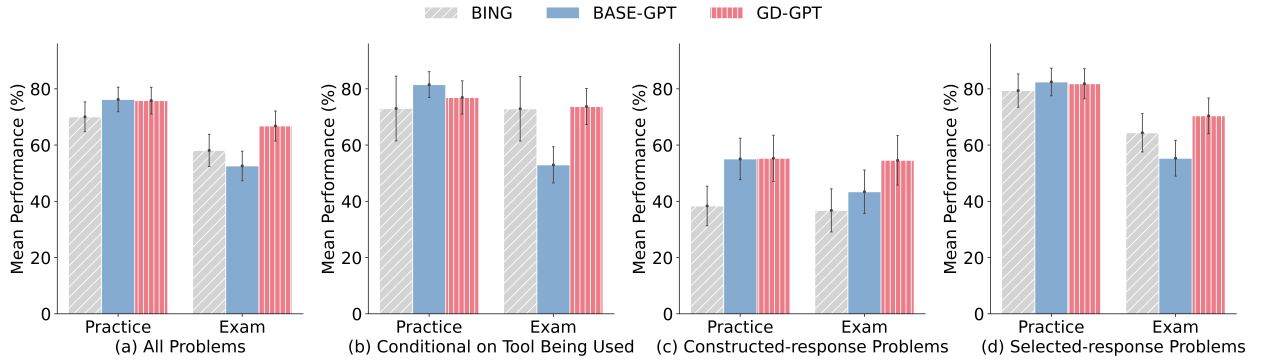
students achieved a mean score of 74.26, using the learning tool on roughly half of the problems. On average, students spent 1.16 minutes per practice problem. During the exam phase, students achieved a mean score of 58.83.

Before the experiment, randomization checks on the pre-survey variables revealed no significant differences across the three groups, as shown in Online Appendix B.

5. Empirical Results

Figure 1 presents the model-free evidence of student learning performance. Plot (a) shows that while both LLMs outperformed BING during practice sessions, only GD-GPT yielded higher exam performance than BING; in contrast, the BASE-GPT group scored lower. Plot (b) reveals that, conditional on tool use, the GD-GPT and BING groups showed similar exam performance, whereas the BASE-GPT group scored significantly lower. Broken down by task format (Plots c-d), the GD-GPT group performed the best on both constructed- and selected-response exam problems. In contrast, BASE-GPT outperformed BING on constructed-response problems but underperformed on selected-response problems.

In sum, model-free evidence suggests that BASE-GPT decreased learning performance (as measured by exam scores), though its harmful effects manifested among selected-response problems but not constructed-response ones. In contrast, our custom GD-GPT enhanced learning performance and was well-received by users, meeting our design objectives.



Note: Whiskers represent 95% confidence intervals

Figure 1 Effects of LLM Chatbots on Student Performance.

5.1. Treatment Effects on Practice and Exam Scores

To statistically test the effects on practice and exam performance, we use the following model:

$$Y_{ij} = \alpha + \beta_1 \times BASE-GPT_{ij} + \beta_2 \times GD-GPT_{ij} + \sigma_j + \delta_{ij} \quad (1)$$

where Y_{ij} denotes the outcome of student i answering problem j , β_1 captures the effect of BASE-GPT compared to BING, β_2 captures the effect of GD-GPT compared to BING, the problem-fixed effect σ_j controls for heterogeneity across problems, and the error term δ_{ij} reflects the idiosyncratic variation in potential outcomes.

Table 2 reports the average treatment effect of providing access to tools (regardless of actual usage) on practice performance and exam performance. Students in the BASE-GPT group scored 7.14 points higher ($p < 0.001$) on practice but 4.18 points lower ($p = 0.026$) on exams than those in the BING condition, suggesting that providing access to BASE-GPT hindered learning. In contrast, the GD-GPT group scored 6.52 and 9.23 points higher on practice and exam problems, respectively, than the BING group ($p < 0.001$ in both cases), suggesting that access to GD-GPT enhanced learning. Compared with BASE-GPT, the GD-GPT group demonstrated similar practice performance ($p = 0.675$) but superior exam performance (13.41 points higher, $p < 0.001$), highlighting the remarkable advantages of our design.⁴

Table 2 Treatment Effects on Practice and Exam Performance

	(1) Practice Score (All Questions)	(2) Exam Score	(3) Practice Score (Selected-response)	(4) Exam Score	(5) Practice Score (Constructed-response)	(6) Exam Score
BASE-GPT	7.140*** (1.51)	-4.180* (1.88)	4.034* (1.79)	-7.718*** (2.28)	17.679*** (2.63)	7.816** (2.80)
GD-GPT	6.522*** (1.56)	9.234*** (1.94)	3.321+ (1.85)	6.543** (2.36)	17.423*** (2.73)	18.372*** (2.90)
Constant	69.464*** (1.12)	57.440*** (1.39)	78.755*** (1.33)	63.719*** (1.69)	37.841*** (1.96)	36.161*** (2.08)
Problem Fixed Effect	Y	Y	Y	Y	Y	Y
Observations	3103	3239	2399	2500	704	739
R^2	0.257	0.117	0.172	0.091	0.154	0.132
GD-GPT – BASE-GPT	-0.618 (1.47)	13.414*** (1.85)	-0.712 (1.74)	14.261*** (2.25)	-0.256 (2.58)	10.556*** (2.76)
Wald test (p-value)	0.675	< 0.001	0.683	< 0.001	0.921	< 0.001

Standard errors in parentheses. + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

5.2. Effects Across Learning Task Formats

To evaluate how the LLM tools' effects vary across task formats, we conducted subsample analyses on two types of problems: constructed-response and selected-response, as shown in Table 2.

During the practice phase, BASE-GPT outperformed BING on both constructed-response problems (by 17.679 points, $p < 0.001$) and selected-response problems (by 4.034 points, $p = 0.024$).

⁴ Appendix C.1 reports treatment effects conditional on actual tool use (see also Fig. b). Although BASE-GPT was the most widely adopted tool, its use was less effective in improving learning outcomes than BING or GD-GPT. BING proved effective when used, but its overall impact was constrained by low adoption. GD-GPT, however, achieved a better balance, offering productivity comparable to BING while maintaining a high usage rate.

However, the effects differed across task formats during the exam phase. Compared to BING, BASE-GPT reduced exam performance on selected-response problems (-7.718 points, $p < 0.001$) but increased performance on constructed-response ones (7.816 points, $p = 0.005$). One explanation for these heterogeneous effects of BASE-GPT is that they reflect how students interacted with the tool differently across problem types. For selected-response problems, students can readily adopt the suggested choice without cognitively processing the tool-generated explanation, thereby bypassing a potential learning opportunity. In contrast, constructed-response problems require students to scrutinize a larger volume of tool-generated information before they can formulate their answers, thereby yielding some learning benefits. For example, in response to the constructed-response problem “*How to address the endogeneity issue to reveal the relationship between IT investment and company profit?*”, BASE-GPT might suggest: “*Use instrumental variables that are correlated with IT investment but not directly with profit, or employ a randomized controlled trial to isolate causal effects.*” Although LLM-generated responses tend to be higher quality and informative than search engine results (as supported by students’ post-survey perceptions in Table 10), students must scrutinize that a randomized controlled trial is impractical in this context, and when students scrutinize the generated answer, they can learn from it, yielding some learning benefits.

GD-GPT outperformed BING during the practice phase on both selected-response problems (by 3.321 points, marginally significant with $p < 0.1$) and constructed-response problems (by 17.423 points, $p < 0.001$). More importantly, it outperformed BING during the exam phase on both problem types: 6.543 points higher on selected-response ones ($p < 0.01$) and 18.372 points higher on constructed-response ones ($p < 0.001$). Compared to BASE-GPT, GD-GPT performed similarly during the practice phase on both types of problems ($\beta = -0.712$, $p = 0.683$ for selected-response problems and $\beta = -0.256$, $p = 0.921$ for constructed-response ones), but significantly better during the exam phase on both problem types (by 14.261 and 10.556 points, respectively, with $p < 0.001$ in both cases).

While we will present a rigorous statistical analysis of the mechanisms driving GD-GPT’s superior performance in Section 6, intuitively, we assert that this advantage stems from a key design feature of GD-GPT: GD-GPT does not provide ready-to-copy answers in either question format, thereby encouraging cognitive processing and active learning. In other words, this design feature forces students to read the explanations and guidance generated by GD-GPT and arrive at their own conclusions from the generated response. Such deeper engagement with the generated response can foster genuine learning, which translates into superior subsequent exam performance.

5.3. Robustness

In this section, we briefly summarize findings from a few robustness checks. Details of these checks are reported in Appendix D. First, our Experiment 1 included an “Ask the bot” button that

automatically pasted the problem into the chat window. This user-interface feature deviated from standard off-the-shelf LLM chatbots and could potentially introduce biases. To rule out this possibility, we removed the button in Experiment 2 while holding all other design elements constant. The findings remained consistent. Second, our main analyses focused on the paper-and-pencil test administered shortly after the practice session. While this assessment captured learning transfer to a new set of problems, it primarily reflected short-term retention. To examine the LLMs’ impact on longer-term learning outcomes, we analyzed students’ performance on subsequent assessments (e.g., exams and assignments) covering the same topics. We found that GD-GPT exhibited persistent performance improvements. Finally, our main analyses used the LLMs’ first response as the source text for measuring critical engagement. Because students may also draw on subsequent LLM responses in their submitted answers, we conducted a robustness check using an alternative measure of critical engagement that used all LLM responses as source text and found similar results.

6. Mechanisms

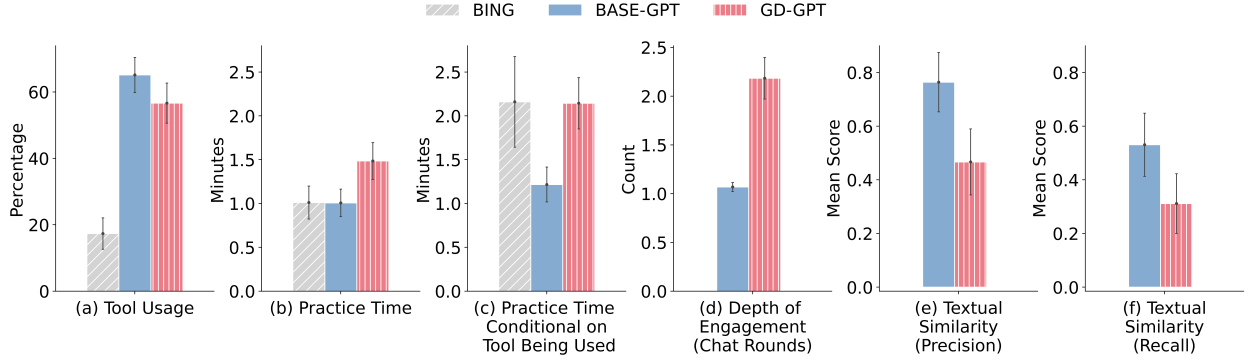
In this section, we unpack the underlying mechanisms for our earlier findings, especially for GD-GPT’s superior performance, by focusing on how students engaged differently with these learning tools and whether such different engagement patterns can explain the differences in learning outcomes.

We will examine three potential mechanisms: *practice time* as a proxy for learners’ effort, *depth of engagement* as measured by the presence of follow-up questions and the number of conversation rounds, and *critical engagement*, which reflects the effort that students devote to scrutinizing the AI outputs, measured by the divergence between generated responses and student-submitted answers. The latter two mechanisms are mainly used to explain the differences between BASE-GPT and GD-GPT. Figure 2 presents the model-free evidence.

6.1. Practice Time

We first examined practice time (i.e., time spent on each practice problem) as a mechanism underlying LLMs’ effects on learning. We began by assessing how the tools differed in their impact on practice time. Column (1) of Table 3 and Fig. 2b show that the BASE-GPT group recorded similar practice time per problem as the BING group ($\beta = 0.03$, $p = 0.585$), whereas the GD-GPT group spent 0.49 minutes (or 48.85%) longer ($p < 0.001$). Intuitively, these differences can be explained by our earlier argument that BASE-GPT encourages direct appropriation of generated answers, which saves time, whereas GD-GPT induces deeper processing of those answers, which takes longer.

We further found that practice time was positively associated with both practice and exam scores. As shown in Columns (2)-(3), each additional minute of practice time was associated with



Note: Whiskers represent 95% confidence intervals

Figure 2 Tool Usage and Underlying Mechanisms.

an average increase of 2.158 points in practice scores ($p < 0.001$) and 1.940 points in exam scores ($p = 0.002$). We then conducted mediation analyses to estimate the effect of GD-GPT on learning performance via practice time and assessed the significance of the conditional indirect effect using bootstrapping. Results indicate that extended practice time is a valid mechanism driving improvements in both practice performance (conditional indirect effect = 1.049, $p < 0.001$) and exam performance (conditional indirect effect = 0.943, $p = 0.003$).

Table 3 Practice Time and Its Relationship to Learning Performance

	(1) Practice Time	(2) Practice Score	(3) Exam Score	(4) Practice Score	(5) Exam Score
BASE-GPT	0.030 (0.05)	7.075*** (1.51)	-3.960* (1.91)	5.350** (1.85)	-2.450 (2.35)
GD-GPT	0.486*** (0.06)	5.473*** (1.57)	6.386** (1.98)	1.384 (1.98)	0.661 (2.48)
Practice Time		2.158*** (0.50)	1.940** (0.63)	0.207 (0.83)	0.503 (1.04)
BASE-GPT $\times$ Practice Time				1.708 (1.08)	-1.467 (1.36)
GD-GPT $\times$ Practice Time				3.378*** (1.00)	4.305*** (1.25)
Constant	0.995*** (0.04)	67.317*** (1.23)	59.097*** (1.55)	69.288*** (1.39)	60.558*** (1.75)
Problem Fixed Effect	Y	Y	Y	Y	Y
Observations	3103	3103	3063	3103	3063
R^2	0.355	0.262	0.119	0.265	0.126
GD-GPT – BASE-GPT	0.456*** (0.05)	-1.602 (1.49)	10.346*** (1.89)	-3.966* (1.90)	3.111 (2.41)
Wald test (p-value)	< 0.001	0.282	< 0.001	0.037	0.196
(GD-GPT – BASE-GPT) $\times$ Practice Time				1.670+ (0.98)	5.772*** (1.24)
Wald test (p-value)				0.088	< 0.001

Standard errors in parentheses. + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

To further understand this mechanism, we examined the marginal effects of practice time on learning performance under different tools by interacting the treatment indicators with students' practice time. As seen in Columns (4)-(5), BASE-GPT produced an immediate increase in practice performance upon chatbot usage ($\beta = 5.350$, $p = 0.004$), but this effect did not grow with practice time ($\beta = 1.708$, $p = 0.113$), suggesting that students might take time to use BASE-GPT only to quickly generate and obtain answers, rather than slowly absorb the generated responses. This pattern also helps explain why BASE-GPT did not improve exam performance. In contrast, GD-GPT had no immediate effect on practice performance ($\beta = 1.384$, $p = 0.485$), but gradually improved it as the practice time grew ($\beta = 3.378$, $p < 0.001$). Moreover, the marginal effect of practice time on exam score was larger under GD-GPT than under BING ($\beta = 4.305$, $p < 0.001$). We see this as an indication that GD-GPT encourages effortful processing of generated responses, which led to learning benefits. In Appendix C.2, we reported similar findings when we examined the effect of practice time *conditional on tool use*.

6.2. Depth of Engagement

Because students learn better if they actively engage with the learning materials (Liu et al. 2017), we investigated depth of engagement, as indicated by follow-up questions and multi-round conversations with LLMs, as another potential mechanism.

Follow-up Question. We first measured the depth of engagement using a binary variable, *Follow-up Question*, which indicates whether a student posted any follow-up questions after receiving the initial LLM response. Our coding of student interactions with LLMs revealed four categories of follow-up questions: (1) concept clarifications, (2) deeper explanation of the question, including the choices in multiple-choice questions, (3) clarification of LLMs' previous responses, and (4) direct inquiry about the correct answer.⁵ Column (1) of Table 4 shows that, compared to BASE-GPT, students in the GD-GPT group were 56.5% more likely to ask follow-up questions.

We further found that follow-up questioning was positively associated with practice and exam performance. On average, students who asked follow-up questions scored 21.686 points higher on practice problems ($p < 0.001$) and 16.240 points higher on exam problems ($p < 0.001$). Mediation analyses confirmed that asking follow-up questions mediated the relationship between GD-GPT and exam performance (conditional indirect effect = 9.138, $p < 0.001$).

Conversation Rounds. The number of conversation rounds is another useful indicator of engagement depth. Results in Table 4 showed that students in the GD-GPT condition, on average, engaged in 1.160 more rounds of conversation per problem than their BASE-GPT counterparts ($p < 0.001$). Meanwhile, each additional round of conversation was associated with a 7.316-point

⁵ We obtained consistent results when excluding "(4) direct inquiry about the correct answer" from the analysis.

Table 4 Follow-up Questions and Conversation Rounds

	(1) Follow-up Question	(2) Practice Score	(3) Exam Score	(4) Rounds	(5) Practice Score	(6) Exam Score
GD-GPT	0.565*** (0.02)	-15.435*** (2.03)	11.915*** (2.91)	1.160*** (0.05)	-11.665*** (1.92)	14.295*** (2.74)
Follow-up Question		21.686*** (2.23)	16.240*** (3.20)			
Rounds					7.316*** (0.92)	5.829*** (1.30)
Constant	0.047*** (0.01)	79.917*** (1.02)	52.050*** (1.49)	1.050*** (0.03)	73.265*** (1.41)	46.726*** (2.02)
Problem Fixed Effect	Y	Y	Y	Y	Y	
Observations	1406	1406	1378	1406	1406	1378
R^2	0.416	0.356	0.203	0.341	0.342	0.199

Notes: We employ a linear probability model for the binary outcome (i.e., *Follow-up Question*) and a linear regression model for the count outcome (i.e., *Rounds*) for ease of interpretation. The results are robust to alternative specifications using logit and negative binomial models.

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

increase in practice scores ($p < 0.001$) and a 5.829-point increase in exam scores ($p < 0.001$). Mediation analyses indicated that conversational rounds positively mediated the relationship between GD-GPT and exam performance (conditional indirect effect = 6.757, $p < 0.001$).

In sum, deeper engagement, as indicated by follow-up questions and a greater number of conversation rounds, appears to be another mechanism underlying GD-GPT’s superior performance relative to BASE-GPT.

6.3. Critical Engagement

As indicated by survey evidence in traditional e-learning, when learners critically evaluate source materials and use them judiciously, they demonstrate better learning performance (Saadé et al. 2012). We therefore investigated critical engagement as a third potential mechanism in LLM-supported learning. We operationalized critical engagement as the dissimilarity between LLM outputs and student-submitted answers. When students adapt and synthesize LLM outputs rather than copying them verbatim, the resulting submitted answers exhibit lower similarity to the original outputs and reflect a higher degree of critical engagement. Specifically, we treated the initial LLM response (which always responds to the original question) as the reference text and measured its semantic and lexical similarity with student-submitted answers.

Semantic Similarity refers to the degree to which two pieces of text share meaning, even if they do not use the same words. We followed a BERT-based approach to measure semantic similarity. We first encoded both sets of documents using Sentence-BERT, a transformer model optimized for semantic similarity tasks. We then computed similarity scores to quantify how similar the encoded student answer was to the LLM response. Column (1) of Table 5 reports the regression results for

the BERT Similarity scores and reveals that students using GD-GPT submitted answers that were semantically more *dissimilar* from LLM responses (-0.125 , $p < 0.001$), indicating a greater degree of critical engagement.

Lexical Similarity refers to how similar two texts are based on shared words and phrases. We measured lexical similarity using ROUGE (Recall-Oriented Understudy for Gisting Evaluation) and BLEU (Bilingual Evaluation Understudy) scores, which capture the degree of overlap between the student’s answer and LLM outputs. ROUGE and BLEU provide complementary perspectives on lexical similarity: ROUGE captures how much of the LLM output appears in the student-submitted answer, while BLEU quantifies the proportion of n-grams in the student’s answer that also appear in the LLM output. Using both metrics allows us to assess not only whether students reproduced LLM outputs but also whether they incorporated and adapted the content in their own answers.⁶

As shown in Column (2), students using GD-GPT produced answers with significantly lower ROUGE scores compared to those using BASE-GPT (-0.220 , $p < 0.001$). This suggests that under GD-GPT, a larger portion of the LLM output was omitted from students’ submitted answers, indicating more critical engagement—students adapted and synthesized the output, rather than copying it verbatim. Column (3) presents results using BLEU: students using GD-GPT submitted answers with significantly fewer consecutive words overlapping with the LLM outputs, indicating greater originality and independence. Combining the semantic and lexical similarity results, we found that GD-GPT prompted students to engage more critically with LLM outputs, demonstrating greater originality and independence in their responses.

Table 5 Similarity Between Student Answers and LLM Response

	(1) BERT Similarity	(2) ROUGE	(3) BLEU
GD-GPT	-0.125*** (0.03)	-0.220*** (0.04)	-0.283*** (0.04)
Constant	0.802*** (0.02)	0.531*** (0.03)	0.327*** (0.02)
Problem Fixed Effect	Y	Y	Y
Observations	361	361	361
R^2	0.194	0.134	0.196

We further found that critical engagement was positively associated with learning outcomes: students whose answers were less similar to LLM outputs during practice achieved higher scores in the exam ($\beta < 0$ across all three similarity measures). Mediation analyses further confirmed that

⁶ Acknowledging that students may ask follow-up questions, we conducted a robust check using aggregate LLM outputs instead of just the initial response as the reference text and found similar patterns. See Appendix D.3 for details.

critical engagement explained performance differences between the two LLMs. Table 6 summarizes the conditional indirect effect for each of the similarity measures, suggesting that boosting critical engagement was one of the mechanisms underlying GD-GPT’s superior learning performance.

Table 6 Summary of Mediation Analyses of Similarity Metrics on Exam Performance

Variable	Conditional Indirect Effect	S.E.	P-value	95% Lower Bound	95% Upper Bound
BERT Similarity	2.383*	1.036	0.021	0.352	4.414
ROUGE	2.348*	1.053	0.026	0.284	4.412
BLEU	3.467*	1.474	0.019	0.579	6.355

7. Additional Analyses

In this section, we extend our main analyses in two ways. First, we conducted post-hoc analyses to explore how the adoption and effects of LLMs vary with several student attributes, including gender, prior experience with LLMs, and course familiarity. Second, we supplemented our main analyses with subjective evaluations of the tools and learning experiences, which we captured during the post-survey.

7.1. Heterogeneity Effects

Gender Differences. We first investigated how the adoption and effects of LLM tools varied by gender. Column (1) of Table 7 shows that female students were 10.6% less likely to use BASE-GPT than male students, consistent with prior evidence that males disproportionately adopt LLMs (Aldasoro et al. 2024). In contrast, female students were 8.7% more likely than males to use GD-GPT. One possible explanation for these gender differences is that BASE-GPT encourages direct copying, thereby increasing the risk of negative judgment from instructors. Given that women tend to exhibit greater risk aversion than men in academic settings (Eckel and Grossman 2008), they may be less inclined to use such tools. GD-GPT substantially reduces these risks and is less often viewed as a “shortcut,” which makes it more attractive to female students. Our results further indicate that gender moderates the impact of GD-GPT on learning outcomes. Female students experienced a 7.736-point greater increase in exam performance than male students when using GD-GPT ($p < 0.1$).

Course Familiarity. We next explored how *Course Familiarity*, defined as a student’s familiarity with the course topics in general, moderated the effects on *Tool Use* and *Exam Score*. As seen in Column (1) of Table 8, neither course familiarity nor its interaction with the LLM (BASE-GPT or GD-GPT) significantly affected students’ tool-use rate. In contrast, we observed significant positive interactions between *Course Familiarity* and LLMs on *Exam Score*, indicating that students with

Table 7 Role of Gender in LLM Adoption and Impact

	(1) Tool Use	(2) Exam Score
BASE-GPT	0.586*** (0.02)	-2.743 (2.47)
GD-GPT	0.372*** (0.03)	5.630* (2.67)
Female	-0.017 (0.03)	-0.949 (2.82)
BASE-GPT $\times$ Female	-0.106** (0.04)	-3.390 (3.79)
GD-GPT $\times$ Female	0.087* (0.04)	7.376+ (3.93)
Constant	0.177*** (0.02)	57.822*** (1.85)
Problem Fixed Effect	Y	Y
Observations	3103	3239
R^2	0.300	0.120

Standard errors in parentheses. + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

greater course familiarity benefit substantially more from accessing the LLMs than from accessing a search engine.

Table 8 Role of Course Familiarity in LLM Adoption and Impact

	(1) Tool Use	(2) Exam Score	(3) Tool Use	(4) Exam Score
BASE-GPT	0.535*** (0.05)	-18.330*** (4.96)	0.626*** (0.05)	-17.754*** (4.72)
GD-GPT	0.345*** (0.05)	-4.529 (5.05)	0.400*** (0.05)	-1.093 (4.85)
Course Familiarity	-0.011 (0.02)	0.124 (1.68)		
BASE-GPT $\times$ Course Familiarity	0.003 (0.02)	6.699** (2.14)		
GD-GPT $\times$ Course Familiarity	0.032 (0.02)	6.550** (2.17)		
Topic Familiarity			-0.004 (0.02)	2.899+ (1.61)
BASE-GPT $\times$ Topic Familiarity			-0.042+ (0.02)	7.073** (2.19)
GD-GPT $\times$ Topic Familiarity			0.007 (0.02)	5.759* (2.29)
Constant	0.195*** (0.04)	57.288*** (3.92)	0.178*** (0.04)	51.763*** (3.54)
Problem Fixed Effect	Y	Y	Y	Y
Observations	3103	3239	3103	3239
R^2	0.294	0.131	0.295	0.137

Standard errors in parentheses. sym+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

A robustness check using topic familiarity (i.e., familiarity with the topics covered in the class session) yielded similar results, as shown in Columns (3)-(4). Taken together, these results suggest

that course familiarity did not affect LLM use, but it magnified the learning benefits associated with LLM access. As a result, tools like GD-GPT disproportionately benefited students who already have some familiarity with the learning topics.

Prior Experience with LLMs. Table 9 reports the heterogeneity analysis based on students’ prior LLM experience. Column (1) shows that students who had used LLMs more frequently previously were significantly more likely to adopt BASE-GPT (0.054, $p < 0.01$), suggesting that prior exposure can lower adoption barriers of LLMs. Column (2) further shows that GD-GPT’s benefits were amplified for students with prior LLM experience: each one-point increase in prior LLM use (on a five-point scale) yielded an additional 3.252-point gain in GD-GPT’s impact on exam performance. This finding can be explained by the fact that students with greater LLM experience are better able to interpret and utilize LLM outputs to support their learning.

Table 9 Role of Prior LLM Experience in LLM Adoption and Impact

	(1) Tool Use	(2) Exam Score
BASE-GPT	0.329*** (0.07)	4.843 (6.77)
GD-GPT	0.454*** (0.06)	-2.975 (6.50)
LLM Experience	0.019 (0.01)	-0.172 (1.23)
BASE-GPT $\times$ LLM Experience	0.054** (0.02)	-2.298 (1.69)
GD-GPT $\times$ LLM Experience	-0.012 (0.02)	3.252* (1.65)
Constant	0.101* (0.05)	58.086*** (4.84)
Problem Fixed Effect	Y	Y
Observations	3103	3239
R^2	0.303	0.121

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

7.2. Subjective Evaluations

In addition to the objective outcomes of learning performance and tool use, we have also captured several subjective evaluations of tools using the post-experiment survey. Appendix E presents the summary statistics and the measurement of these items.

We first analyzed results on three dimensions of perceived information quality, including *Information Amount*, *Information Relevance*, and *Information Diversity*. We also investigated the effects on *Perceived Usefulness*, which assessed the extent to which the students believed that the tool enhanced task performance (Hess et al. 2014), *Intention to Use*, which reflected students’ readiness to adopt the learning tool in the future (Venkatesh and Davis 1996), and *Intrinsic Motivation*,

which captured students’ inherent willingness and effort to learn (Park et al. 2019, Yi and Davis 2003). Compared to those using BING, students using BASE-GPT reported higher *Information Amount* (0.228, $p = 0.083$), *Information Relevance* (0.299, $p = 0.02$), and *Information Diversity* (0.254, $p = 0.04$). Students using GD-GPT perceived an even higher level of information quality than those using BASE-GPT across all three dimensions: *Information Amount* (0.285, $p = 0.028$), *Information Relevance* (0.380, $p = 0.003$), and *Information Diversity* (0.317, $p = 0.009$). These findings on information quality supported our design goal of making GD-GPT a valuable learning tool. We further find that students perceived BASE-GPT (0.394, $p < 0.001$) and GD-GPT (0.674, $p < 0.001$) as significantly more useful than search engines, showed a significantly higher intention to use BASE-GPT (0.453, $p = 0.001$) and GD-GPT (0.576, $p < 0.001$), and reported higher intrinsic motivation under BASE-GPT (0.330, $p = 0.01$) and GD-GPT (0.409, $p = 0.003$).

Overall, these findings corroborated the objective results showing that LLM tools were far more likely to be used by students than BING and revealed a highly favorable perception of GD-GPT. Interestingly, while students rated GD-GPT higher in information quality and usefulness, they used BASE-GPT more frequently. This apparent “contradiction” suggests that students may struggle to resist BASE-GPT, as its lower effort demands make it appealing even though they acknowledge it is less effective for learning. This observation aligns with prior reports noting that “students worry they learn nothing from generative AI, but they use it anyway (Ross 2025).”

Table 10 Perceived Information Quality and Intention to Use

	(1) Information Amount	(2) Information Relevance	(3) Information Diversity	(4) Perceived Usefulness	(5) Intention To Use	(6) Intrinsic Motivation
Base-GPT	0.228 ⁺ (0.13)	0.299* (0.13)	0.254* (0.12)	0.394*** (0.12)	0.453*** (0.13)	0.330* (0.13)
GD-GPT	0.513*** (0.14)	0.679*** (0.14)	0.571*** (0.13)	0.674*** (0.12)	0.576*** (0.14)	0.409** (0.13)
Constant	3.334*** (0.10)	3.362*** (0.10)	3.331*** (0.09)	3.313*** (0.09)	3.326*** (0.10)	3.389*** (0.10)
Observations	231	231	231	231	231	231
R^2	0.089	0.127	0.117	0.149	0.136	0.090
GD-GPT – BASE-GPT	0.285* (0.13)	0.380** (0.13)	0.317** (0.12)	0.279* (0.11)	0.123 (0.13)	0.079 (0.12)
Wald Test (p-value)	0.028	0.003	0.009	0.014	0.33	0.527

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

7.3. Effects on Cognitive Load

We also used the post-survey to investigate LLMs’ impact on students’ cognitive processes. We measured cognitive load using four constructs (Krieglstein et al. 2023): *Intrinsic Load* captured the cognitive demands imposed by the complexity of the practice problems. *Extraneous Load* reflected

the cognitive demands caused by irrelevant or poorly presented information. *Germane Load* measured the constructive cognitive effort devoted to schema building and real learning. Germane load reflects higher-order cognitive activities, such as critical thinking and problem-solving, which are essential for long-term retention and mastery of the material. Finally, *Cognitive Overload* captured the extent to which the cognitive demands exceed working memory capacity (Krieglstein et al. 2023, Paas et al. 2010).

Table 11 reveals that, compared to BING, both BASE-GPT and GD-GPT significantly reduced intrinsic load ($p < 0.001$) and extraneous load ($p < 0.001$), indicating that the LLMs required fewer cognitive resources than the search engine. Relatedly, students were less likely to experience cognitive overload when using LLMs compared to search engines ($p < 0.001$ and $p < 0.05$, respectively). These findings reflect that LLMs are much easier to use than search engines, which explains their higher usage rates.

Students reported a higher germane load under BASE-GPT and GD-GPT ($p < 0.001$ in both cases), which suggested enhanced schema building and deeper learning under the two LLMs (Krieglstein et al. 2023). Moreover, students using GD-GPT exhibited a significantly higher level of germane load than those using BASE-GPT (0.316, $p = 0.004$). The latter finding is consistent with the objective result that GD-GPT produced better learning outcomes than BASE-GPT. On the other hand, although students reported higher germane load under BASE-GPT than under BING, the objective exam performance revealed the opposite: BASE-GPT resulted in lower learning outcomes. One way to explain this “discrepancy” is that students overestimated BASE-GPT’s learning benefits, which gave them the illusion of learning while their real learning suffered (Deslauriers et al. 2019).

Table 11 Effects on Cognitive Load

	(1) Cognitive Overload	(2) Intrinsic Load	(3) Extraneous Load	(4) Germane Load
Base-GPT	-0.641*** (0.14)	-0.695*** (0.14)	-0.934*** (0.14)	0.580*** (0.11)
GD-GPT	-0.353* (0.14)	-0.579*** (0.15)	-0.760*** (0.15)	0.896*** (0.12)
Constant	3.428*** (0.10)	3.592*** (0.11)	3.515*** (0.11)	3.089*** (0.09)
Observations	231	231	231	231
R^2	0.135	0.161	0.205	0.258
GD-GPT – BASE-GPT	0.289* (0.13)	0.116 (0.14)	0.174 (0.14)	0.316*** (0.11)
Wald Test (p-value)	0.03	0.405	0.214	0.004

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

8. Discussion

Due to the ability of LLM chatbots to generate fluent, contextually relevant, and on-demand responses, students of all ages have rapidly adopted these tools to help with their school work; however, concerns are growing that such use may enable students to bypass learning processes and ultimately hinder, rather than enhance, learning. Motivated by limited evidence on when and how LLM use impacts learning, we conducted field experiments to examine how access to LLMs affects learning outcomes, and whether these effects depend on LLM design, the nature of learning tasks, and other contextual factors. Our randomized field experiments indicate that standard LLMs are not uniformly beneficial for learning; their effects can be either positive or negative, depending crucially on the chatbot design and the nature of learning tasks. Specifically, relative to search engines, granting students access to standard LLMs hurts learning, but this negative effect emerges only for selected-response problems, not for constructed-response problems. More importantly, guided-discovery LLMs, which are designed to guide students to arrive at their own answers, can yield significantly better learning outcomes for both problem types and enhance learning overall. Guided-discovery LLMs are more effective because they can encourage students to spend more time on learning and engage with LLMs more deeply and critically.

We also find that the benefits of guided-discovery LLMs vary by several learner characteristics. Interestingly, although female students adopt standard LLMs at a lower rate than their male counterparts—consistent with prior evidence of a gender gap in AI adoption—they adopt guided-discovery LLMs at a higher rate and benefit more from them, effectively reversing this gap. We also find that students with stronger prior knowledge and greater experience using LLMs benefit more from guided-discovery LLMs.

Our study also confirms the presence of an “illusion of learning” when students use standard LLMs: although actual learning performance is low, their self-reported learning is high. Moreover, we find evidence of “over-reliance” on standard LLMs—students evaluate guided-discovery LLMs, including their intention to use them, more favorably, even though they use standard LLMs more frequently. Together, these findings underscore a fundamental challenge in promoting learning-oriented LLMs: students’ reliance and adoption decisions are likely driven more by convenience and perceived learning than by actual learning effectiveness.

8.1. Contributions to the Literature

Our study contributes to multiple literature streams. First, it enriches the emerging literature on human-LLM interaction by advancing our understanding of how LLMs impact learning. We demonstrate that LLMs, at least the standard ones, are not uniformly beneficial for learning; instead, their effects are highly contingent upon design, task, and individual characteristics. We note that

the effects of LLMs on learning differ markedly from their effects on tool-assisted task performance: although students with access to standard LLMs demonstrate improved task performance during practice, they achieve lower learning outcomes on subsequent exams. Recognizing this divergence is crucial when designing LLMs intended to support not only task performance but also learning and human development. Consistent with this distinction, our findings show that LLMs are more effective learning tools when students are required to exert greater cognitive effort, such as working on constructed-response rather than selected-response tasks, and when LLMs are designed to guide human discovery rather than substitute it. By identifying the interaction design principles and task conditions that make LLMs a more effective learning tool, this study advances our understanding of when and how human-LLM interactions can foster genuine learning.

Second, this study contributes to technology-mediated learning by advancing understanding of how LLMs can function as effective learning support tools rather than mere answer generators. Our findings show that the learning benefits of LLMs critically depend on whether core pedagogical principles, particularly guided discovery, are embedded into their interaction design. Importantly, we demonstrate that these pedagogical principles can yield substantial learning benefits when implemented through prompt-based interaction design, without requiring domain-specific tutoring logic, handcrafted scaffolds, or costly model fine-tuning. This prompt-based approach is theoretically grounded and highly scalable, offering a practical pathway for integrating learning theory into LLM-supported learning environments and for designing learning-oriented LLMs.

Third, we extend guided discovery theory by deepening our understanding of its underlying mechanisms using fine-grained behavioral trace data from human-LLM interaction. Prior research on guided discovery has largely relied on final learning outcomes; Our field experiments capture moment-by-moment learner behaviors, allowing us to examine how guided discovery reshapes learners' behaviors and cognitive processes to produce superior learning outcomes. Specifically, we show that guided discovery increases learning time, depth of engagement, and critical engagement, and that these mechanisms help explain how this long-standing learning theory translates into meaningful human-LLM interaction patterns that support genuine learning. However, this pattern coexists with evidence that female learners adopt guided-discovery LLMs at higher rates, indicating that motivation to engage with guided discovery may be heterogeneous across learners and warrants further investigation. Moreover, we show that guided-discovery LLMs are more effective for learners with stronger prior knowledge, suggesting that guided discovery may impose greater cognitive demands. Future research should therefore explore how guided-discovery designs can be adapted to better support learners who are less prepared or less confident.

8.2. Practical Implications

This study offers several actionable insights for educators, students, administrators, and learning system designers to better understand the impacts of LLMs on learning and to effectively harness them to advance learning.

First, we show that standard LLMs are not well-suited for learning for several reasons. (a) Unfettered access to standard LLMs can undermine learning outcomes. (b) Because standard LLMs are optimized to deliver fluent, ready-made responses, they do not inherently incorporate pedagogical principles necessary to realize the full learning potential of LLMs. (c) Moreover, they generate an illusion of learning, whereby learners report high perceived learning despite poorer objective performance. (d) Finally, by providing immediate, complete answers, standard LLMs foster over-reliance, encouraging cognitive offloading rather than active engagement.

Instead, responsible LLM designs should focus on supporting and augmenting learning processes rather than bypassing them. Our findings suggest that educators and administrators should encourage the adoption of LLMs designed around pedagogical principles, such as guided discovery. Luckily, prompt-based interaction design offers a practical and effective way to integrate pedagogical principles into LLMs. Educators and institutions can leverage this approach to create customized LLM chatbots and guide students toward using these learning-oriented designs rather than standard LLMs. More recently, major LLM providers have begun to integrate learning-oriented design elements resembling guided discovery, although empirical evidence on their learning effectiveness remains limited. Meanwhile, students should be encouraged to incorporate guided-discovery principles into their interactions with LLMs, even when they have access only to standard LLMs. For example, rather than asking for direct answers, students can prompt LLMs to help break down problems, provide relevant background knowledge, pose follow-up questions, and critically evaluate LLM outputs instead of accepting them at face value.

Moreover, educators and students should be trained to recognize that the fluent and seemingly plausible answers generated by LLMs can create an illusion of learning. The availability of instant, ready-made responses may further foster over-reliance, even among students who are aware of the potential drawbacks of relying on LLM-generated answers. Accordingly, students should be encouraged to recognize these blind spots and to take active steps to mitigate them.

Our studies also suggest that instructional designers should design learning assignments to prevent cognitive offloading. Task design, fully under instructors' control, critically shapes LLM effectiveness, especially when standard LLM access is unavoidable. Instructors should, for example, avoid relying on selected-response questions for formative learning activities and instead emphasize constructed-response tasks that require students to generate and articulate their own reasoning. More broadly, instructional strategies that promote sustained cognitive engagement may further

reduce offloading and mitigate the potential harms of irresponsible LLM use. For instance, instructors can ask students to document their interactions with LLMs and to reflect on what they learned, how the LLM supported their thinking, and where its responses fell short.

Finally, our findings highlight a trade-off between pedagogical rigor and popular appeal in the design of educational LLMs. Although guided-discovery LLMs are rated more favorably, their higher cognitive demands lead to lower adoption than standard LLMs. This tension suggests that effective LLM design must balance learning effectiveness with usability, and that optimal designs may vary by context and learner characteristics.

8.3. Limitations and Future Research

This study is subject to several limitations, which also point to important directions for future research. First, our study is conducted among college students and with a limited set of task formats. Future research is needed to assess whether the observed effects generalize to other educational levels, such as K-12, and to alternative task designs, including project-based and collaborative tasks. Second, the study focuses on the role of LLMs in formative practice assignments. More research is needed to generalize our findings to other learning processes, such as in-class instruction, self-directed learning, and collaborative learning. Third, although the guided discovery design improves overall learning performance, it appears to favor students with strong prior knowledge, raising concerns about the potential widening of achievement gaps. Future studies should investigate design modifications that improve accessibility for lower-prepared students. Finally, we operationalize guided discovery using zero-shot prompt engineering to prioritize ease of implementation and scalability. More advanced LLM design strategies, such as fine-tuning, retrieval-augmented generation, and agentic workflows, may further enhance guided discovery and therefore warrant future investigation.

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Online Appendices for “Is ChatGPT a Boon or a Bane for Learning? Experimental Evidence Across Task Formats and Chatbot Designs”

Appendix A presents system interface screenshots for the assisted practice phase, along with the prompts used in the BASE-GPT and GD-GPT conditions. Appendix B reports randomization checks, comparing student characteristics across the three experimental groups. Appendix C shows the effects conditional on tool use, including impacts on learning performance and practice time. Appendix D provides robustness checks, including long-term learning outcomes, an alternative interface design tested in the second experiment, and alternative textual similarity measures. Appendix E lists the post-survey questionnaire items and summary statistics.

Appendix A: System Interface for Assisted Practice

A.1. Screenshots

Figure EC.1 presents a screenshot of the system interface for the BING group.

Question 4 out of 10:

The experimental design is typically divided into two main types: between-subjects design and within-subjects design. In a between-subjects design, participants are randomly assigned to different experimental groups, and each group receives a different treatment or condition. In a within-subjects design, the same group of participants receives all experimental treatments at different time points. Regarding the Wilcoxon Rank Sum Test for experimental data, which of the following statements is incorrect?

A) The procedures for conducting the Wilcoxon Rank Sum Test are the same for both types of experiments.

B) We can use the Wilcoxon Test for hypothesis testing regardless of the distribution of the experimental results.

C) For within-subjects experiments, only some of the data may be meaningful for the Wilcoxon Test.

D) Outliers have a relatively small impact on this method.

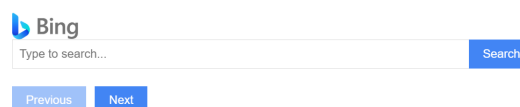


Figure EC.1 A Screenshot of the Practice Interface for the BING Group

Figure EC.2 presents a screenshot of the system interface design for the LLM chatbot (i.e., BASE-GPT and GD-GPT) groups.

Question 4 out of 10:

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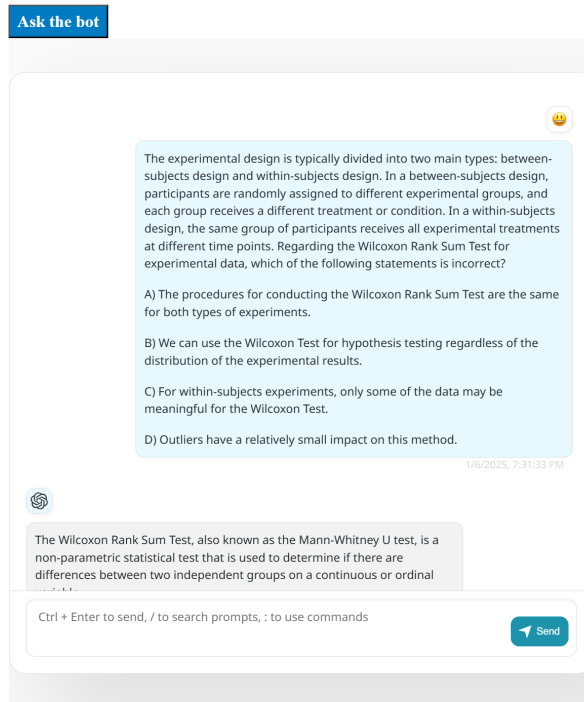


Figure EC.2 A Screenshot of the Practice Interface for the LLM Groups

A.2. System Prompts

The following paragraph describes the system prompt we deployed for the BASE-GPT chatbot.

You are a teaching assistant to help students with practice questions for a course. The student will be presented with a question on the left side of the screen and will answer it there. You are shown on the right side of the screen as a helper.

We additionally incorporated the guidelines informed by the guided discovery principle in the GD-GPT system prompt as follows.

You are a teaching assistant to help students with practice questions for a course. The student will be presented with a question on the left side of the screen and will answer it there. You are shown on the right side of the screen as a helper.

You will follow the "guided discovery" principle in offering assistance to the students. Specifically, you will:

- 1) help the students break down the problem and structure the thought process, if possible.
- 2) provide essential background knowledge, especially in areas where students are prone to make mistakes.
- 3) allow students to independently arrive at the answer by applying concepts. You may offer more assistance if the student asks for follow-up help.

What you should not do:

- 1) you should not provide the final answer to the question like "the correct answer is ...".
- 2) you should not comment on the correctness of each choice, if multiple choices are provided. For example, you should not say "Option A is incorrect".

To summarize, you will help students answer the question by providing background knowledge and structuring their thought process, but not revealing the answer directly.

Appendix B: Randomization Checks

To verify the validity of randomization, we compared the participant characteristics across the Search Engine, BASE-GPT, and GD-GPT conditions. The results showed that the three groups were balanced in terms of *Gender* (the proportion of Female = 0.42, 0.41, and 0.43, respectively; $p = 0.980$), *Age* (21.92, 22.18, and 22.08; $p = 0.865$), and the proportion of *Domestic* students (0.97, 0.96, and 0.96; $p = 0.927$). We also found no significant differences in perceived *Social Status* (6.88, 7.03, and 6.96; $p = 0.824$), personal *Learning Goal* (3.22, 3.20, and 3.11; $p = 0.610$), frequency of *Search Engine* Use (4.33, 4.38, and 4.39; $p = 0.924$), or frequency of *LLM Use* (4.03, 4.04, and 4.00; $p = 0.961$). Similarly, no differences emerged in students' *IT Self-efficacy* (3.48, 3.53, and 3.60; $p = 0.633$), *Course Familiarity* (2.18, 2.23, and 2.18; $p = 0.905$), or *Topic Familiarity* (2.07, 2.07, and 2.02; $p = 0.903$). Taken together, these results confirmed that the randomization procedure successfully produced comparable groups.

Table EC.1 Randomization Checks

Variable	Search Engine (N=96)	BASE-GPT (N=104)	GD-GPT (N=96)	p-value (F-test)
Female	0.42	0.41	0.43	0.980
Age	21.92	22.18	22.08	0.865
Domestic	0.97	0.96	0.96	0.927
Social Status	6.88	7.03	6.96	0.824
Learning Goal	3.22	3.20	3.11	0.610
Search Engine Use	4.33	4.38	4.39	0.924
LLM Use	4.03	4.04	4.00	0.961
IT Self-efficacy	3.48	3.53	3.6	0.633
Course Familiarity	2.18	2.23	2.18	0.905
Topic Familiarity	2.07	2.07	2.02	0.903

Appendix C: Treatment Effects Conditional on Tool Use

Model-free evidence suggests that students used BING for an average of 18.37% of practice problems, compared with 70.52% and 58.10% for BASE-GPT and GD-GPT, respectively, indicating students' strong preference for LLM chatbots over the search engine. To statistically investigate the relative appeal of different tools, we examined the likelihood of tool use by a linear probability model. As shown in Table EC.2 Column (1), students were significantly more likely to use BASE-GPT (by 54.2%, $p < 0.001$) and GD-GPT (by 41.3%, $p < 0.001$) compared to BING. The relatively high usage rate of GD-GPT (although slightly lower than BASE-GPT) suggested that our customized chatbot was well-received by students.

C.1. Treatment Effects on Learning Performance Conditional on Tool Use

Next, we estimated the conditional treatment effects by restricting the sample to the practice problems where students used the assigned tool and their corresponding matched exam problems. As shown in Table EC.2, on average, students using BASE-GPT achieved practice scores 7.822 points higher ($p = 0.004$) compared to those using BING, and such increase likely stemmed from the provision of direct answers to the BASE-GPT group. However, this advantage in the practice came at a cost of learning in the exam: these BASE-GPT students on average performed 16.658 points worse on exam problems ($p < 0.001$), suggesting that the use of BASE-GPT undermined learning. In contrast, students using GD-GPT showed modest, though not statistically significant, improvements in both practice ($\beta = 4.695$, $p = 0.063$) and exam scores ($\beta = 4.460$, $p = 0.226$) relative to those using BING. Compared to BASE-GPT, while GD-GPT had a lower usage rate possibly due to its lack of direct answers and a slightly lower practice score (3.127, $p = 0.063$), it yielded a significant 21.118-point increase in exam performance ($p < 0.001$).

Table EC.2 Effect on Practice and Exam Performance Conditional on Tool Use

	(1) Tool Use	(2) Practice Score (Conditional on Tool Use)	(3) Exam Score (Conditional on Tool Use)
BASE-GPT	0.542*** (0.02)	7.822** (2.69)	-16.658*** (3.62)
GD-GPT	0.413*** (0.02)	4.695+ (2.74)	4.460 (3.68)
Constant	0.171*** (0.01)	73.095*** (2.42)	69.863*** (3.26)
Problem Fixed Effect	Y	Y	Y
Observations	3103	1577	1549
R^2	0.293	0.288	0.183

Standard errors in parentheses

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

These results suggest that while BASE-GPT was the most popular tool, it was not as productive as BING or GD-GPT when students really used it for learning. BING remained effective when students actually use it, but was adopted far less frequently than the LLM chatbots, largely reducing its practical effectiveness. GD-GPT, however, balanced these competing demands more effectively, yielding high-performing learning outcomes comparable to Bing while maintaining a substantial level of user appeal.

C.2. Effects of Practice Time Conditional on Tool Use

Practice time represents the duration that students spent working on a practice problem. Conditional on tool use, students using BASE-GPT spent significantly less time per problem than BING users (0.629 fewer minutes, $p < 0.001$) and GD-GPT users (0.795 fewer minutes, $p < 0.001$). In contrast, students using GD-GPT spent a comparable amount of time as those using BING ($\beta = 0.166, p = 0.177$). We further investigated the marginal effect of practice time conditional on tool use and found consistent results: the marginal effect of practice time was significantly higher when students used GD-GPT (see Columns (4) and (5) of Table EC.3). As practice time acts as a proxy of student attention and engagement level, our findings indicate that GD-GPT enhanced learning efficiency by encouraging deeper engagement and more productive cognitive processing during practice than BASE-GPT.

Table EC.3 Impact of Practice Time Conditional on Tool Use

	(1) Duration	(2) Practice Score	(3) Exam Score	(4) Practice Score	(5) Exam Score
BASE-GPT	-0.629*** (0.12)	8.488** (2.71)	-16.076*** (3.65)	5.608 (4.09)	-18.455*** (5.52)
GD-GPT	0.166 (0.12)	4.519+ (2.74)	4.307 (3.68)	-4.045 (4.33)	-5.310 (5.83)
Practice Time		1.059+ (0.58)	0.927 (0.78)	-1.045 (1.41)	-1.236 (1.90)
BASE-GPT $\times$ Practice Time				0.639 (1.58)	0.180 (2.13)
GD-GPT $\times$ Practice Time				3.916* (1.55)	4.384* (2.09)
Constant	1.929*** (0.11)	71.053*** (2.66)	68.064*** (3.59)	75.708*** (3.85)	72.885*** (5.18)
Problem Fixed Effect	Y	Y	Y	Y	Y
Observations	1577	1577	1549	1577	1549
R^2	0.370	0.290	0.183	0.296	0.189

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Appendix D: Robustness Checks

D.1. Robustness of Treatment Effects: Post-experimental Learning Performance

To assess whether learning gains persist beyond the experimental session, we examined students' performance on subsequent course assessments covering the same topics. Specifically, we collected performance data from post-experiment homework assignments and quizzes, and from exam questions (for courses with final exams) related to the topics practiced during the field experiment. To rigorously measure the effect of tool-assisted practice on post-experimental performance, each post-experiment assessment question was manually evaluated by trained research assistants for topical overlap with in-experiment practice problems and matched when sufficient overlap was identified. Of the 3,279 practice problems used in the experiment, 1,826 could be matched to at least one subsequent assessment question. Each of the 1,826 practice problems was matched to an average of 2.64 follow-up questions (ranging from one to six), yielding a total of 4,847 post-experimental observations. These data allow us to estimate the long-term effects of the experimental treatments on learning performance beyond the immediate practice and exam phases.

Table EC.4 Effect on Post-experimental Learning Performance

	(1) All	(2) Topics in Selected- Response Practices	(3) Topics in Constructed- Response Practices
Base-GPT	0.794 (1.07)	-0.374 (1.24)	6.485*** (1.83)
GD-GPT	6.148*** (1.11)	5.580*** (1.28)	8.911*** (1.90)
Constant	76.536*** (0.79)	80.317*** (0.92)	58.086*** (1.36)
Problem Fixed Effect	Y	Y	Y
Observations	4847	4023	824
R^2	0.181	0.135	0.229

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

As shown in Column (1) of Table EC.4, students who used GD-GPT achieved a significant improvement in post-experimental performance, scoring on average 6.148 points higher on subsequent assessments covering the same topics practiced during the experiment than the BING group. In contrast, students using Base-GPT did not differ significantly from those using BING. These results indicate that the learning gains associated with GD-GPT persisted beyond the experimental session, whereas the negative short-term effects observed for Base-GPT attenuated over time. We further examined the heterogeneous effects across different types of questions used during experimental practice. For topics practiced via selected-response questions, as shown in Column (2), Base-GPT was associated with a slight 0.374-point decline in subsequent performance (n.s.), while GD-GPT continues to yield a substantial 5.580-point increase ($p < 0.001$). For topics practiced via constructed-response questions, as shown in Column (3), both Base-GPT and GD-GPT generated persistent learning gains, with performance increases of 6.485 and 8.911 points, respectively.

Post-experimental learning performance suggested that learning gains from guided-discovery LLM use were durable, whereas the adverse effects of standard LLM use attenuated over time. One possible explanation is that students using BASE-GPT, upon recognizing their knowledge gaps from exam performance during the experiment, might have exerted additional learning effort before the subsequent assessments, such as final examinations. However, this remains speculative, since these post-experimental out-of-classroom behaviors were not observable in our study. Importantly, the positive treatment effect of GD-GPT extended beyond the short experimental term and was associated with robust and sustained improvements in long-term learning outcomes.

D.2. Robustness of Interface Design in the Second Experiment

To ensure that our findings were not driven by a specific interface design, we modified the assisted-practice interface for the second experiment. Specifically, as shown in Figure EC.2, our first experiment featured an Ask the bot button that automatically pasted practice problems into the chat window, after which students received LLM responses and could pose follow-up questions.

In the second experiment, we removed the “Ask the bot” function and allowed students in both the Base-GPT and GD-GPT conditions to interact freely with the LLMs. The second experiment included 287 students, who completed 4,592 practice problems and an equal number of exam problems. Results in Table EC.5 confirmed that the results of the second experiment were consistent with those from the first, indicating that our findings were not artifacts of the automatic copy-pasting of practice questions.

Table EC.5 Robustness of the Second Experimental Period

	(1) Practice Score (All Questions)	(2) Exam Score	(3) Practice Score (Selected-response)	(4) Exam Score	(5) Practice Score (Constructed-response)	(6) Exam Score
BASE-GPT	11.340*** (1.18)	-4.418** (1.45)	12.257*** (1.83)	-9.440*** (2.21)	9.811*** (0.79)	3.952*** (1.11)
GD-GPT	10.383*** (1.21)	10.000*** (1.48)	9.355*** (1.87)	6.667** (2.26)	12.097*** (0.81)	15.556*** (1.13)
Constant	67.540*** (0.85)	52.339*** (1.04)	71.505*** (1.32)	60.430*** (1.60)	60.932*** (0.57)	38.853*** (0.80)
Problem Fixed Effect	Y	Y	Y	Y	Y	Y
Observations	4592	4592	2870	2870	1722	1722
R^2	0.060	0.051	0.027	0.021	0.190	0.110

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

D.3. Robustness of Textual Similarity: Using Aggregated Bot Responses

Table EC.6 demonstrates the similarity between student answers and aggregated bot responses. The results remained consistent: students using GD-GPT submitted answers that were more dissimilar to bot responses compared to those using BASE-GPT.

Table EC.6 Alternative Measures of Textual Similarity Using Aggregated Bot Responses

	(1) BERT Similarity	(2) ROUGE Precision	(3) ROUGE Recall	(4) BLUE
GD-GPT	-0.126*** (0.03)	-0.225*** (0.04)	-0.288*** (0.04)	-0.285*** (0.04)
Constant	0.803*** (0.02)	0.792*** (0.03)	0.523*** (0.03)	0.326*** (0.02)
Problem Fixed Effect	Y	Y	Y	Y
Observations	361	361	361	361
R^2	0.198	0.181	0.201	0.202

Standard errors in parentheses. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Appendix E: Questionnaire Items

Table EC.7 Summary Statistics of Post-Experimental Survey (N=231 students)

Variable	Description	Mean	SD	Min	Max
Information Amount	Amount of information provided by this tool is very high, from Strongly Disagree (1) to Strongly Agree (5)	3.61	0.80	1	5
Information Relevance	Relevance of information provided by this tool is very high, from Strongly Disagree (1) to Strongly Agree (5)	3.69	0.84	1	5
Information Diversity	Diversity of information provided by this tool is very high, from Strongly Disagree (1) to Strongly Agree (5)	3.58	0.84	1	5
Perceived Usefulness	The extent to which a person believes that using a particular system or tool will enhance their task performance or outcomes.	3.68	0.76	1	5
Intention to Use	An individuals conscious plan or readiness to adopt, engage with, and continue using a particular system, technology, or tool in the future.	3.69	0.85	1	5
Cognitive Overload	The extent to which the cognitive demands exceed the limited capacity of working memory.	3.06	0.89	1	5
Intrinsic Load	The cognitive load imposed by the learning tasks complexity.	3.13	0.95	1	5
Extraneous Load	The cognitive load imposed by the way information is presented or by irrelevant distractions.	2.91	0.98	1	5
Germane Load	The constructive cognitive effort that promotes learning via schema construction.	3.60	0.79	1	5
Learning Motivation	The willingness and effort a learner invests in acquiring knowledge or skills.	3.65	0.82	1	5

Table EC.8 Questions For the Post-experimental Survey**Perceived Usefulness**

The next few questions are about your perception of the learning tool. Please answer how much you agree or disagree with each statement.

This tool enables me to complete the practice problems more quickly.

Using this tool enhances my effectiveness in completing the practice problems.

This tool enables me to learn the relevant knowledge more quickly.

Using this tool enhances my effectiveness in learning the relevant knowledge.

Intention to Use

Please answer how much you agree or disagree with each statement.

Assuming I had access to this learning tool, I intend to use it in learning.

Given that I had access to this learning tool, I predict that I would use it in learning.

Cognitive Overload

How do you think about the information provided by the learning tool?

The information was very overwhelming.

The information was too complex for me to understand.

The information was too much to process at once.

The information was excessive and hard to concentrate on.

Intrinsic Load

The next few questions are about your perceptions of the practice problems you have taken. Please answer how much you agree or disagree with each statement.

The practice problems were difficult to understand.

The practice problems included much complex information.

Extraneous Load

The next few questions are about your perception of the learning tool. Please answer how much you agree or disagree with each statement.

The design of the learning tool was inconvenient.

The design of the learning tool made it difficult to find relevant information quickly.

Because of the design of the learning tool, I had the impression that I could not concentrate on the learning content.

Germane Load

When I used the learning tool to help solve the practice problems,

I actively reflected upon the topics covered by practice problems.

I made an effort to understand the topics covered by practice problems.

I achieved a comprehensive understanding of the topics covered by practice problems.

I can apply the knowledge that I acquired through the learning tool quickly and accurately.

I was able to expand my prior knowledge with the topics covered by practice problems.

Learning Motivation

How do you think about the topic you have learnt today?

It is fun and interesting to learn [Topic name].

I enjoy learning [Topic name].

It is an exciting thing to learn [Topic name].

I am highly interested in learning [Topic name].